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Get Started with ABAP Cloud

Get the most important information about ABAP Cloud in one page.

Getting to Know ABAP Cloud

Recommended Readings

General Introduction

The following chapters give you a conceptual introduction to ABAP Cloud and its background without diving deep into the technological details.

- Why ABAP Cloud?: A short introduction as to why ABAP Cloud is important.
- ABAP Cloud Development Model: Gives you a brief overview of the main points that are relevant to understanding ABAP Cloud.

Key Concepts

The core concepts sum up the technological basics upon which ABAP Cloud development is built. They require a basic understanding of how the ABAP ecosystem works.

- Cloud-Optimized ABAP Language: Introduces the role of the ABAP language and ABAP language versions in ABAP Cloud.
- Public Released APIs: Introduces the concepts of released APIs, and why they are important when developing in ABAP Cloud.
- Development Scenarios: Gives you an overview of the end-to-end use cases that you can cover with ABAP Cloud.

ABAP Cloud Development and Delta Scenarios

The ABAP Cloud development and delta scenarios assist you in understanding the possibilities for developing with ABAP Cloud:

- Development scenarios: Development scenario describe end-to-end development use case like developing a SAP Fiori app from scratch.
- Delta scenarios: Delta scenario add functioality or features to already existing develop scenarios like for example extending an existing SAP Fiori app with extension fields.

The scenarios provide a holistic perspective for the main relevant use cases. They guide the development of your applications and services in ABAP Cloud along pre-defined paths.

Scenario Overview

ABAP Cloud supports the following scenarios:

Development Scenarios

Development Scenario	Characteristics	Domain-Specific Implementation
Transactional Apps	Transactional use cases require apps and services that run operations such as read, create, update, or delete operations on the data set (OLTP). The architectural separation of concerns between the domain-specific implementation and the business service exposure lets you expose the same data model for analytical and transactional apps and services. Transactional apps are exposed with OData and SAP Fiori elements.	• Implemented with: ABAP RESTful Application Programming Model (RAP) • Domain-Specific Model: RAP business object • Domain-specific Logic: ABAP (mainly Entity Manipulation Language), Core Data Services (CDS) For details, see Transactional Apps.
Analytical Apps	Analytical use cases involve analyzing and evaluating multidimensional data models to derive real-time data-driven business decisions. With analytical apps, you can create data models to analyze business data in embedded or cross-system setups, and to visualize the data in dashboards or as part of SAP Fiori apps. Analytical apps are exposed with InA and SAP Fiori elements or other analytical clients.	• Implemented with: Core Data Services • Domain-Specific Model: Analytical Provider (multidimensional data model with cube and dimensions) • Domain-specific Logic: ABAP, Core Data Services (CDS)

Development Scenario		Characteristics	Domain-Specific Implementation
Integration Services	Process Integration	Process integration use cases can range from data exchange across system boundaries to trigger follow-on actions with events when a value in an app is changed. With process integration, the communication between the communication partners for point-to-point integration can be bidirectional, meaning that information is exchanged in both directions. There are both synchronous and asynchronous process integration patterns, for example event-based integration follows an asynchronous approach. Process integration is exposed as an integration service with the respective protocol that is included in a communication scenario and communication arrangement.	• Generally implemented with (depending on protocol): ABAP Core Data services/ ABAP • Business Events and OData Web APIs: Implemented with the ABAP RESTful Application Programming Model (RAP) • Domain-Specific Logic: ABAP, Core Data Services For more information, see Integration Services.
	Data Integration	Data integration use cases exchange data between two or more parties without being part of specific and predefined business process. Data integration is exposed as an integration service with the SQL binding that is included in a communication scenario and communication arrangement.	• Implemented with: Core Data Services • Domain-Specific Logic: SQL For more information, see Integration Services.

Delta Scenarios

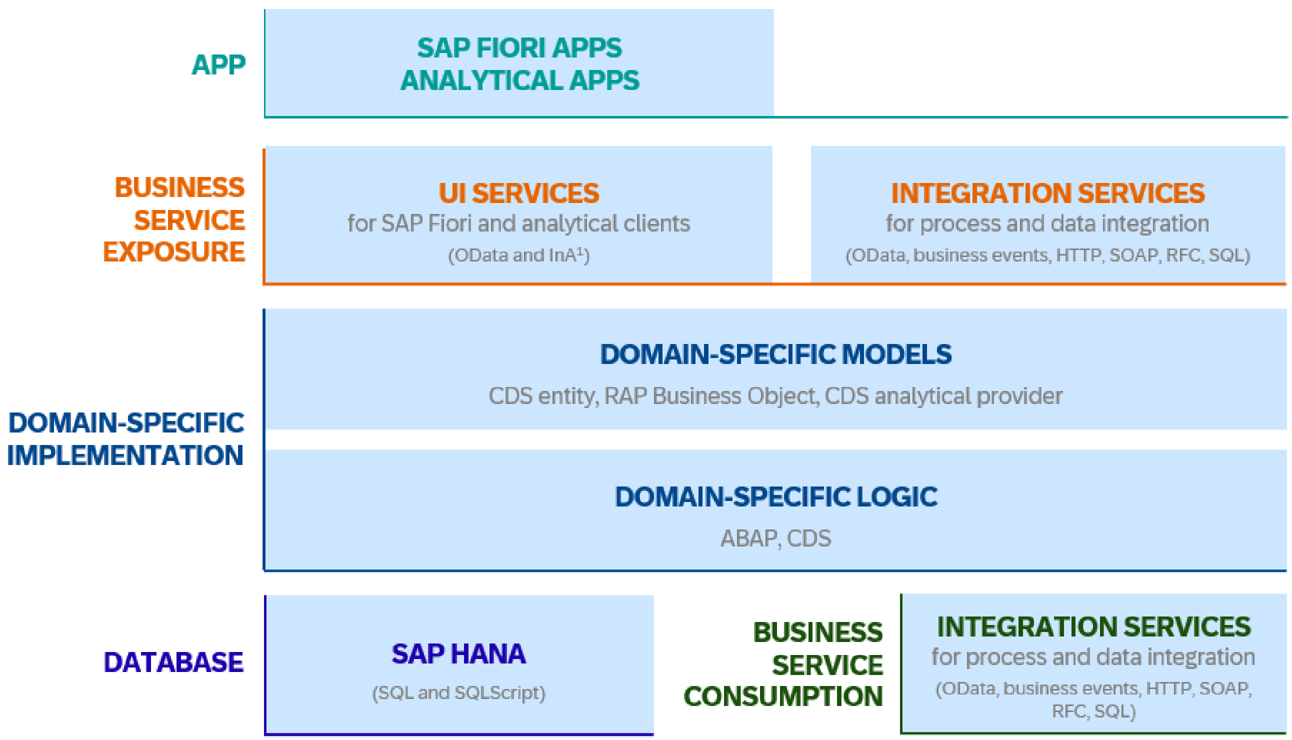
Delta Scenario	Characteristics	Can be used for	Implementation with
Extensibility	Extensibility enables you to add fields or behavior extensions or data model extensions to exisiting SAP Fiori apps or services on multiple levels. Typical use cases are adding fields or data model extenions to apps delivered by SAP.	• Transactional Apps • Analytical Apps • Events	• Implemented with: ABAP, Core Data Services (CDS) For more information, see https://www.sap.com/documents/2022/10/52e0cd9b-497e-0010-bca6-c68f7e60039b.html.

Background Information about ABAP Cloud Architecture

ABAP Cloud is based on a model-driven architecture approach that focuses on improving development efficiency through standardization and formalization of the programming model and the tooling environment to ensure effieciency and scalability.

Programming models generally define the design-time software architecture with specific technologies, concepts, and development objects. It essentially defines a standard architecture for app and service development, from the database to the business service exposure. For more background information about the model-driven architecture, see Conceptual Background: Model-Driven Architecture.

The model-driven architecture approach is adpoted by all ABAP Cloud wizards that generate ABAP Cloud development scenarios. The architecture is based on a three-tier architecture with a database tier, an application tier and a presentation tier. The data access is implemented either with a SAP HANA database, where data is queried with SQL and SQLScript, or a business service consumption, where the data source originates from a different system. The domain-specific implementation defines the design-time for the data models with ABAP Core Data Services(CDS) and the domain-specific logic that is executed at runtime. The domain-specific logic is implemented either with CDS or ABAP. The following graphic shows an abstract design-time architecture for transactional apps, analytical apps and integration services with the aspects that they have in common:



- **Database**: The data access tier is defined by the database or a business consumption model. through database or business service consumption. The data layer for the use cases consists either of an SAP HANA database, where data is queried with SQL and SQLScript, or a business service consumption where the data source originates from a different system.
- **Domain-specific Implementation**: The domain-specific implementation defines the application tier of ABAP Cloud consisting of the domain-specific data models and the domain-specific logic. The domain-specific model defines the structure of the data model where the data is read and the domain-specific logic implements the processing of the data, or in case of a transactional app or service, the respective behavior for a RAP business object.
- **Business Service Exposure**: Defines an additional abstraction layer between the client and the service exposure. The business service exposure defines the protocol in which a data model and its implementation is exposed for consumption by different clients. UI services are exposed in OData for transactional SAP Fiori Apps and InA for analytical SAP Fiori apps. Integration services are exposed in various protocols that are available.

- **App**: SAP Fiori apps require specific development objects and role information, so that the tiles can be exposed on a SAP Fiori launchpad.

Get Started with ABAP Development Tools for Eclipse and other Tools

Each persona in ABAP Cloud has their dedicated tooling environment that is adapted to their specific requirements.

ABAP Development Tools for Eclipse

Developers require a full-fledged integrated development environment that supports the end-to-end development process and to increase developer productivity as much as possible.

The ABAP development tools for Eclipse (ADT) are the integrated development environment on the well-known Eclipse platform for all standard ABAP development, quality assurance, and supportability tasks in ABAP Cloud. ADT offers a modern development toolset with many advantages including full support for development for the domain-specific implementation with ABAP Core Data Services (CDS), ABAP-Managed Database Procedures (AMDP), and the ABAP RESTful Application Programming Model (RAP), or ABAP Analytics.

For information about how to get started with ABAP Development Tools for Eclipse, see Basic Tutorial.

Tools for Key Users and Enterprise Analytics Users

Key users are the business department experts or implementation partner experts who create so-called last mile adaptations with high product skills but with limited technical/coding skills.

Key users use key user tools. Key user tools are web-based Fiori apps that don't require any configuration of the development environment. They provide easy, guided extensibility that's suitable for key users, based on a no code/low code approach. With key user tools, you can implement, for example, adaptations for analytics, forms UI adaptations, custom fields, or logic extensions.The availabilit of key user tools depends on the specific solution scope. To get an overview, see:

- SAP S/4HANA Cloud: Key User Extensibility
- SAP S/4HANA: Key User Extensibility
- SAP BTP, ABAP environment: Extensibility

Tools for Administrators

Administrators have technical tasks related to setting up the system. Administrators take care to set up the monitoring or communication management in ABAP Cloud.

ABAP Cloud provides a dedicated tooling set for administrators that allows them to accomplish their tasks. The Fiori apps that are available for administrators include apps for the following:

- Identity and Access Management (IAM): IAM apps secure access to the solution for business users.
- Communication management: Communication management is the basis for integration services to enable data exchange across system boundaries.
- Security: Security apps ensure a safe and GDPR-compliant system setup.
- Technical monitoring: With apps for technical monitoring, administrators can monitor the system workload and resource consumption in the ABAP and HANA systems, ABAP work processes, and the performance impact of SQL statements.

Get Started with Development

You can try out one of the main ABAP Cloud development scenarios to get to know more about the different possibilities in ABAP Cloud. You can find the most important information with links to more details to get you started on the following pages:

- Transactional Apps
- Analytical Apps
- Integration Services